

YELLOWSTONE, WY, USA

WMO: 726664

Lat: 44.544N

Lon: 110.421W

Elev: 2388

StdP: 75.74

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94173

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-24.7	-31.0	0.3	-27.8	-27.5	0.4	-24.0	5.8	-5.4	5.2	-4.7	0.5	310	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	25.1	11.1	23.5	10.8	22.1	10.4	13.4	20.3	12.5	19.7	11.7	19.0	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	10.9	15.2	9.7	10.0	14.8	8.6	9.3	13.9	44.9	20.1	42.1	19.5	39.7	19.0	17.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.8	4.2	DB	-35.0	27.2	1.9	1.1	-36.4	28.0	-37.5	28.7	-38.5	29.4	-39.9	30.2	(4)
(5)			WB	-34.0	15.4	1.4	1.1	-35.0	16.2	-35.8	16.8	-36.6	17.4	-37.6	18.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-10.7	-9.9	-5.8	-1.6	3.6	8.7	13.3	12.1	7.8	1.3	-4.6	-9.5	(6)	
	DBStd	9.57	6.12	6.04	5.06	4.10	3.47	3.48	2.36	2.48	3.41	4.38	5.74	7.00	(7)	
	HDD10.0	3712	643	557	490	348	201	63	4	11	81	271	438	607	(8)	
	HDD18.3	6529	901	791	748	598	458	289	154	194	316	529	687	864	(9)	
	CDD10.0	226	0	0	0	0	1	25	108	74	14	0	1	2	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	145	0	0	0	0	0	11	90	39	3	0	0	1	(12)	
	CDH26.7	4	0	0	0	0	0	0	4	1	0	0	0	0	(13)	
(14) Wind	WSAvg	1.3	0.9	1.2	1.3	1.3	1.2	1.4	1.4	1.4	1.3	1.3	1.3	1.1	(14)	
(15) Precipitation	PrecAvg	504	49	36	40	39	51	57	39	44	42	34	43	40	(15)	
	PrecMax	712	122	135	116	78	119	138	105	108	101	154	99	101	(16)	
	PrecMin	346	6	10	10	11	11	11	0	6	3	2	7	2	(17)	
	PrecStd	81	25	22	21	19	26	29	26	24	25	25	22	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	5.2	9.6	12.9	19.0	24.9	27.2	26.3	23.3	16.8	15.9	17.2	(19)	
		MCWB	1.9	0.0	2.4	3.8	8.1	10.9	11.8	11.0	10.2	6.8	3.1	3.7	(20)	
	2%	DB	1.4	2.1	6.8	10.4	16.2	22.5	25.7	24.4	21.2	13.8	8.5	7.9	(21)	
		MCWB	-0.4	-0.6	0.6	2.8	6.9	10.1	11.4	10.5	9.4	5.9	2.0	0.4	(22)	
	5%	DB	-1.0	0.2	4.1	8.0	13.7	20.5	24.2	22.8	19.4	11.3	5.5	2.3	(23)	
		MCWB	-2.3	-1.5	-0.2	1.7	5.9	9.4	11.3	10.3	8.7	4.8	1.0	-0.5	(24)	
	10%	DB	-2.6	-1.2	2.0	5.7	11.3	18.2	22.6	21.3	17.3	8.7	3.0	-0.8	(25)	
		MCWB	-3.8	-2.6	-1.1	1.0	5.0	8.8	11.0	10.2	8.0	3.6	-0.1	-1.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	0.9	2.8	5.0	9.3	12.7	14.8	14.5	12.8	8.5	4.4	3.6	(27)	
		MCDB	8.1	2.5	7.2	11.7	16.1	21.5	21.9	20.6	18.9	14.8	10.3	16.7	(28)	
	2%	WB	-0.3	-0.2	1.5	3.2	7.9	11.1	13.6	13.1	11.1	6.8	2.7	0.8	(29)	
		MCDB	2.0	1.8	5.2	8.6	14.4	19.3	20.5	19.7	17.5	12.2	7.2	5.6	(30)	
	5%	WB	-2.1	-1.3	0.3	2.3	6.6	10.1	12.8	12.1	10.0	5.5	1.4	-0.3	(31)	
		MCDB	-1.1	0.2	3.5	6.8	12.7	18.4	20.1	19.0	16.8	10.2	4.9	2.6	(32)	
	10%	WB	-3.6	-2.6	-0.7	1.3	5.4	9.2	12.0	11.3	8.9	4.1	0.3	-1.8	(33)	
		MCDB	-2.6	-1.2	2.0	5.0	10.5	17.0	19.6	18.3	15.5	8.1	2.9	-0.4	(34)	
(35) Mean Daily Temperature Range	MDBR		11.0	12.7	13.4	12.3	12.1	15.0	17.5	17.4	15.8	11.6	10.8	10.0	(35)	
	5% DB	MCDBR	10.7	12.8	15.2	15.9	17.5	19.9	20.4	20.5	20.4	16.4	12.3	10.4	(36)	
		MCWBR	8.4	9.9	9.9	9.0	9.1	9.1	9.1	9.7	10.4	9.4	7.2	6.0	(37)	
	5% WB	MCDBR	9.8	11.2	13.1	13.8	16.2	17.7	17.7	17.6	17.8	15.1	10.3	8.2	(38)	
		MCWBR	7.8	8.7	8.7	8.1	8.8	8.6	8.7	9.3	9.8	9.2	6.5	5.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.220	0.230	0.240	0.294	0.292	0.294	0.301	0.308	0.281	0.262	0.232	0.218	(40)	
	taud		2.369	2.330	2.353	2.297	2.505	2.572	2.584	2.529	2.616	2.616	2.522	2.396	(41)	
	Ebn at Noon		935	981	1011	970	978	972	962	946	955	938	918	903	(42)	
	Edh at Noon		73	93	105	123	103	97	95	97	81	70	63	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.67	2.61	3.84	5.09	5.61	6.63	7.12	5.98	4.62	2.81	1.73	1.29	(44)	
	RadStd		0.21	0.29	0.44	0.46	0.65	0.61	0.37	0.33	0.44	0.27	0.20	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page